

MTD 367 iOS Application Development

- ~~Lesson #1 - Building Apps That Matter and Swift~~
- ~~Lesson #2 - Dealing With Swift Types And SwiftUI~~
- ~~Lesson #3 - Doubling Down On SwiftUI~~
- ~~Lesson #4 - All To Do With SwiftData~~
- **Lesson #5 - Streaming Multimedia Swiftly (Today)**
- Lesson #6 - Putting The Map On Your App (15/9/2025)

? Questions About Lesson #4

- Basics Of SwiftData
 - Why Persist data?
 - What Is SwiftData?
 - How SwiftData Works?
 - SwiftData Vs CoreData

? Questions About Virtual Lab #4

Improve your To-Do List app!

- Use a gesture to mark a task as done
- Display done in a separate list
- Display done and in-progress tasks separately
- Use a gesture to switch views between done and in-progress

✓ Today's Agenda

- ☐ Learn how to show images, play audio or play video in a Swift app

Swift Multimedia Features

Media Type	Details
 Video	- Play local videos with <code>VideoPlayer</code> and <code>Bundle</code> - Stream videos from internet URLs
 Audio	- Play local audio with <code>AVAudioPlayer</code> - Stream audio from internet URLs using <code>AVPlayer</code>
 Images	- Load images from assets with <code>Image("...")</code> - Load images from internet using <code>AsyncImage</code>
 Media Picker	- Select photos or videos from device library using <code>PhotosPicker</code>

XCode



Hands-On #6

Improve your To-Do List app!

- Play motivational audio when task is completed!

✓ Summary

- ✓ Learn how to show images, play audio or play video in a Swift app

Virtual Lab #5 (13/9/2025)

Improve your To-Do List app!

- Let User add their own images and videos in their tasks.

? Questions?

I'd love to hear how the class went or how I can improve.

Feel free to email me at:

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 All course materials are available at:

www.github.com/iluzdaf/mtd367

Appendix

Playing Video

```
import SwiftUI
import AVKit

struct VideoView: View {
    let url = URL(string: "https://example.com/video.mp4")!
    var body: some View {
        VideoPlayer(player: AVPlayer(url: url))
            .frame(height: 300)
    }
}
```

Also works with local files

```
let localURL = Bundle.main.url(forResource: "myVideo", withExtension: "mp4")!  
VideoPlayer(player: AVPlayer(url: localURL))
```

Playing Audio

```
import AVFoundation

let url = URL(string: "https://example.com/audio.mp3")!
let player = AVPlayer(url: url)
player.play()
```

Playing local audio files

```
import AVFoundation

var player: AVAudioPlayer?

func playLocalAudio() {
    if let url = Bundle.main.url(forResource: "sound", withExtension: "mp3") {
        do {
            player = try AVAudioPlayer(contentsOf: url)
            player?.play()
        } catch {
            print("Audio error: \(error)")
        }
    }
}
```

Showing Images

Displaying Images From Assets:

```
Image("photoName")  
  .resizable()  
  .scaledToFit()
```

Loading Images From the Internet using AsyncImage

```
AsyncImage(url: URL(string: "https://example.com/photo.jpg")) { image in
    image.resizable().scaledToFit()
} placeholder: {
    ProgressView()
}
.frame(width: 300, height: 300)
```


Loading Media from Device

Local Images / Videos / Audio:

- Bundle resources
 - Put your media files in Xcode → Resources folder
- File Manager
 - Access user documents folder
- Photo Picker (for photos or videos)

```
import PhotosUI

struct PhotoPickerExample: View {
    @State private var selectedImage: UIImage?

    var body: some View {
        PhotosPicker(
            selection: .constant(nil),
            matching: .images
        ) {
            Text("Pick a Photo")
        }
        .onChange(of: selectedImage) { newImage in
            // do something with newImage
        }
    }
}
```